

Multiple Choice

1. **Answer:** A

Options: A) RbBr only; B) RbBr, Rb, Br₂, and Rb₂Br; C) RbBr and Rb₂Br only; D) RbBr, Rb, and Br₂; E) Rb and Br₂ only

Note: Correct option: A - RbBr only

2. **Answer:** D

Options: A) The hydration energies of cerium ions and sulfate ions are very high.; B) The heat of solution of cerium(III) sulfate is isothermic.; C) The solubility of cerium(III) sulfate increases with temperature.; D) The heat of solution of cerium(III) sulfate is exothermic.; E) The hydration energies of cerium ions and sulfate ions are very low.

Note: Correct option: D - The heat of solution of cerium(III) sulfate is exothermic.

3. **Answer:** A

Options: A) 4×10^{39} ; B) 6×10^{36} ; C) 7×10^{37} ; D) 5×10^{39} ; E) 9×10^{41}

Note: Correct option: A - 4×10^{39}

4. **Answer:** B

Options: A) 8.67 g; B) 3.82 g; C) 5.11 g; D) 3.00 g; E) 4.45 g

Note: Correct option: B - 3.82 g

5. **Answer:** C

Options: A) 26.3 g mol⁻¹; B) 500 g mol⁻¹; C) 212 g mol⁻¹; D) 0.122 g mol⁻¹; E) 1.22 g mol⁻¹

Note: Correct option: C - 212 g mol⁻¹

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $8.61 \times 10^{-3} \text{kg m}^5 \text{s}^{-2} \text{mol}^{-2}$ '.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $\text{GeH}_4 < \text{SnH}_4 < \text{PbH}_4 < \text{SiH}_4 < \text{CH}_4$ '.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** $5.665 \times 10^{17} \text{ sec}^{-1}$. *Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.*

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

11. **Answer:** b and c are Hermitian. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key